# BACHELOR OF COMPUTER APPLICATION

## SEMESTER 1

# FUNDAMENTALS OF MATHEMATICS

# Unit 12

# Integration Part I

## Table of Contents


| SL.No | Topic | | Fig No / Table / Graph | SAQ / Activity | Page No |
|---|---|---|---|---|---|
| 1 | Introduction | | - | - | 4 - 5 |
| | 1.1 | Learning Objectives | - | - | |
| 2 | Integration | | - | - | 6 - 9 |
| | 2.1 | Basic Concepts of Integration | - | - | |
| | 2.2 | Indefinite Integrals | - | - | |
| | 2.3 | Definite Integrals | - | - | |
| | 2.4 | Fundamental Theorem of Calculus | - | - | |
| | 2.5 | Techniques of Integration | - | - | |
| | 2.6 | Applications of Integration | | | |
| 3 | Integration of Common Functions | | - | - | 10 - 11 |
| | 3.1 | Basic Integration Formulas | - | - | |
| | 3.2 | Integration of Polynomials | - | - | |
| | 3.3 | Integration of Logarithmic Functions | - | - | |
| | 3.4 | Integration of Exponential Functions | - | - | |
| 4 | Integration of Circular and Inverse Circular Functions | | - | - | 12 - 14 |
| | 4.1 | Integration of Circular (Trigonometric) Functions | - | - | |
| | 4.2 | Integration of Inverse Circular (Inverse Trigonometric) Functions | - | - | |
| | 4.3 | Techniques for Integrating Trigonometric and Inverse Trigonometric Functions | - | - | |
| 5 | Integration of Simple Functions Using Substitution | | - | - | 15 - 18 |
| | 5.1 | Understanding Substitution in Integration | - | - | |
| 6 | List of Integration Formulas | | - | - | 19 - 20 |
| 7 | Summary | | - | - | 21 - 22 |
| 8 | Self-Assessment Questions | | - | 1 | 23 - 24 |

| | | | | |
|---|---|---|---|---|
| 9 | Terminal Questions | - | - | 25 |
| 10 | Answers | - | - | 26 |

## 1. INTRODUCTION

Welcome to Unit 12: Integration Part I! In our last adventure through Unit 11, we unpacked the dynamic world of derivatives, focusing on their real-world applications such as calculating rates of change, understanding the behavior of increasing and decreasing functions, and the ever-important tangents and normals. We also navigated through the fascinating landscape of maxima and minima, using both the first and second derivative tests to determine the peaks and troughs of function curves. Moreover, we tackled problems that solidified our understanding of derivatives in actual scenarios and explored the crucial concepts of finding absolute maximum and minimum values over closed intervals.

Now, as we pivot to Unit 12, we dive into the complementary world of integration. Why should you care about integration, you ask? Well, integration is not just about solving mathematical puzzles; it's about understanding accumulation. From calculating areas under curves to determining the total distance traveled by a car under varying speeds, integration helps us aggregate quantities that change continuously—making it a cornerstone of not only mathematics but also physics and engineering. This unit will guide you through the basic to advanced concepts of integration, enriching your mathematical toolkit significantly.

How do we go about mastering integration? We'll start with the basics: understanding the integral as the inverse process of differentiation. You'll learn to solve indefinite integrals, which will help you grasp the concept of antiderivatives. Then, we'll explore definite integrals, where you'll learn to calculate the exact area under curves between two points—equipping you with the skills to apply integration to real-world problems. Alongside these, we'll delve into the Fundamental Theorem of Calculus, a principle that beautifully connects differentiation and integration. By the end of this unit, you'll not only be able to solve integrals but also apply them in practical and theoretical contexts, preparing you for more complex scenarios and further studies in calculus. Let's integrate our knowledge and discover how the accumulation of small parts builds a whole!

## 1.1. Learning Objectives

*At the end of this unit, you will be able to:*

- *Identify the basic notation and symbols used in integration, such as the integral sign and dx.*
- *Explain the concept of integration as the inverse process of differentiation.*
- *Compute the definite and indefinite integrals of simple functions using basic integration formulas.*
- *Differentiate between the methods of substitution and integration by parts to determine the appropriate technique for solving a given integral.*

INSPIRED BY LIFE

# 2. INTEGRATION

Integration is a fundamental concept in calculus that complements differentiation. While differentiation deals with finding the rate of change of a function, integration is concerned with finding the total accumulation of quantities. In essence, integration is the inverse process of differentiation.

## 2.1. Basic Concepts of Integration

Integration is the process of finding the integral of a function. The integral represents the accumulation of quantities and can be thought of as the area under the curve of a function.

**Notation**

The integral of a function $f(x)$ with respect to $x$ is denoted by:

$$\int f(x)\,dx$$

The symbol $\int$ represents the integral sign, $f(x)$ is the integrand, and $dx$ indicates the variable of integration.

## 2.2. Indefinite Integrals

An indefinite integral, also known as an antiderivative, represents a family of functions whose derivative is the original function. If $F(x)$ is an antiderivative of $f(x)$, then:

$$\int f(x)\,dx = F(x) + C$$

Here, C is the constant of integration, accounting for the fact that the derivative of a constant is zero.

**Example:**

Find the indefinite integral of $f(x) = 2x$.

**Solution:**

$$\int 2x\,dx = x^2 + c$$

## 2.3. Definite Integrals

A definite integral calculates the net area under the curve of a function between two specified points, $a$ and $b$. It is represented as:

$$\int_a^b f(x)\,dx$$

The definite integral has both an upper limit $b$ and a lower limit $a$. The result is a number representing the accumulated quantity over the interval $[a,b]$.

**Example:**

Find the definite integral of $f(x) = 2x$ from $x = 1$ to $x = 3$.

**Solution:**

$$\int_1^3 2x\,dx = [x^2]_1^3$$

$$\int_1^3 2x\,dx = (3)^2 - (1)^2 = 9 - 1 = 8$$

## 2.4. Fundamental Theorem of Calculus

The Fundamental Theorem of Calculus links differentiation and integration, showing that they are inverse processes.

**Part 1:** Connection Between Derivative and Integral

If $F(x)$ is an antiderivative of $f(x)$, then:

$$\frac{d}{dx}\left(\int_a^x f(t)\,dt\right) = f(x)$$

This part states that differentiation and integration are inverse operations.

**Part 2:** Evaluation of Definite Integrals

If $F(x)$ is an antiderivative of $f(x)$, then:

$$\int_a^b f(x)\,dx = F(b) - F(a)$$

This part allows us to evaluate the definite integral by finding the difference between the values of the antiderivative at the upper and lower limits.

## 2.5. Techniques of Integration

There are several techniques for finding integrals, including:

1. **Basic Integration Rules:** These include the power rule, sum rule, and constant multiple rules, analogous to differentiation rules.
2. **Substitution:** A method that simplifies the integrand by changing variables.

3. **Integration by Parts:** A technique based on the product rule for differentiation.
4. **Partial Fractions:** Decomposing a rational function into simpler fractions that are easier to integrate.

## 2.6. Applications of Integration

Integration has numerous applications in various fields:

1. **Area Under a Curve:** Calculating the area between a function and the x-axis over a given interval.
2. **Accumulated Quantity:** Finding the total accumulated change over time, such as distance travelled given a velocity function.
3. **Physics and Engineering:** Computing quantities like work, center of mass, and electric charge distribution.
4. **Economics:** Determining consumer and producer surplus, and other accumulated measures.

**Example:**

Find the area under the curve $f(x) = x^2$ from $x = 0$ to $x = 2$.

**Solution:**

$$\int_0^2 x^2\,dx = \left[\frac{x^3}{3}\right]_0 = \frac{2^3}{3} - \frac{0^3}{3} = \frac{8}{3}$$

A car accelerates from rest at a constant rate of $3m\ /\ s^2$. How far does the car travel in the first 5 seconds?

**Solution:**

The acceleration $a(t)$ is given by $3m|s^2$. The velocity $v(t)$ is the integral of acceleration:

$$v(t) = \int a(t)\,dt = \int 3\,dt = 3t + c$$

Since the car starts from rest, $v(0)$, so $c = 0$:

$$v(t) = 3t$$

The distance $s(t)$ is the integral velocity:

$$s(t) = \int v(t)\,dt = \int 3t\,dt = \frac{3t^2}{2} + c$$

Since the car starts from rest, $s(0) = 0$, so $c = 0$:

$$s(t) = \frac{3t^2}{2}$$

To find the distance travelled in the first 5 seconds:

$$S(5) = \frac{3(5)^2}{2} = \frac{3 \cdot 25}{2} = 37 \cdot 5\ meters$$

The car travels 37.5 meters in the first 5 seconds.

Integration, as the inverse process of differentiation, plays a crucial role in calculus. It allows us to find the total accumulation of quantities, such as areas under curves and the total distance travelled. Understanding the relationship between differentiation and integration, along with mastering various integration techniques, is essential for solving a wide range of mathematical and real-world problems.

# 3. INTEGRATION OF COMMON FUNCTIONS

Integration is a fundamental concept in calculus that involves finding the antiderivative of a function. This process is essential for solving various problems in mathematics, physics, engineering, and other fields. This guide covers the integration of common functions, including basic integration formulas and specific examples involving polynomials, logarithms, and exponentials.

## 3.1. Basic Integration Formulas

The following are some essential integration formulas that form the foundation for integrating more complex functions:

1. Power Rule:

   $\int x^n \, dx = \frac{x^{n+1}}{n+1} + c$ for $n \neq -1$

2. Exponential Function

$$\int e^x \, dx = e^x + c$$

3. Logarithmic Function

$$\int \frac{1}{x} dx = \ln|x| + c$$

## 3.2. Integration of Polynomials

Polynomials are among the most straightforward functions to integrate. The integration process involves applying the power rule to each term of the polynomial.

**Example**

Integrate $f(x) = 3x^2 - 5x + 4$

**Solution:**

$$\int (3x^2 - 5x + 4) \, dx = \int 3x^2 \, dx - \int 5x \, dx + \int 4 \, dx$$

Using the power rule for each term:

$$\int 3x^2 \, dx = 3\int x^2 \, dx = 3 \cdot \frac{x^3}{3} = x^3$$

$$\int 5x \, dx = 5\int x \, dx = 5 \cdot \frac{x^2}{2} = \frac{5x^2}{2}$$

$$\int 4 \, dx = 4x$$

Combining the results:

$$\int (3x^2 - 5x + 4)\, dx = x^3 - \frac{5x^2}{2} + 4x + c$$

## 3.3. Integration of Logarithmic Functions

Integrating logarithmic functions often involves using substitution or recognizing the integral's form. The most common logarithmic function integrated is $\frac{1}{x}$.

## 3.4. Integration of Exponential Functions

Exponential functions are frequently encountered in various fields, and their integration is straightforward due to the constant ratio of change property.

Integrating common functions, including polynomials, logarithms, and exponentials, involves using basic integration formulas and techniques such as substitution and integration by parts. Mastery of these methods is essential for solving more complex integrals and understanding the applications of integration in various fields. By practicing these examples and applying the fundamental integration rules, one can develop a strong foundation in integral calculus.

# 4. INTEGRATION OF CIRCULAR AND INVERSE CIRCULAR FUNCTIONS

Integration of circular (trigonometric) and inverse circular (inverse trigonometric) functions is a crucial aspect of calculus, enabling us to solve various problems involving periodic phenomena and angles. This guide provides an overview of the techniques and examples related to the integration of these functions.

## 4.1. Integration of Circular (Trigonometric) Functions

Trigonometric functions such as sine, cosine, and tangent have well-known integrals that are frequently used in various fields, including physics, engineering, and signal processing.

**Basic Integrals of Trigonometric Functions**

1. Sine Function

$$\int sin(x)\, dx = -cos(x) + c$$

2. Cosine Function

$$\int \cos(x)\, dx = \sin(x) + c$$

3. Tangent Function

$$\int \tan(x)\, dx = -\ln|\cos(x)| + c$$

4. Cotangent Function

$$\int \cot(x)\, dx = \ln|\sin(x)| + c$$

5. Secant Function

$$\int \sec(x)\, dx = \ln|\sec(x) + \tan(x)| + c$$

6. Cosecant Function

$$\int \operatorname{cosec}(x)\, dx = -\ln|\operatorname{cosec}(x) + \cot(x)| + c$$

**Example:**

Find the integral of $f(x) = sin(x)\, cos(x)$.

**Solution:**

Using the trigonometric identity $sin(x)\, cos(x) = \frac{1}{2} sin(2x)$:

$$\int sin(x) \cdot cos(x) = \int \frac{1}{2} sin(2x)\, dx = \frac{1}{2} \int sin(2x)\, dx$$

Let $u = 2x$, then $du = 2\, dx$ or $dx = \frac{du}{2}$:

$$= \frac{1}{2}\int sin(u) \cdot \frac{du}{2}$$

$$= \frac{1}{4}\int sin(u)\, du$$

$$= \frac{1}{4}(-cos(u)) + c$$

$$\int sin(x) \cdot cos(x) = -\frac{1}{4}cos(2x) + c$$

Find the integral of $f(x) = sec^2(x)$

**Solution:**

Using the known integral of the secant squared function:

$$\int sec^2(x)\, dx = tan(x) + c$$

## 4.2. Integration of Inverse Circular (Inverse Trigonometric) Functions

Inverse trigonometric functions, such as arcsine, arccosine, and arctangent, are used to find angles given the values of trigonometric functions. Their integrals are essential in various applications, including integration by parts and substitution.

**Basic Integrals of Inverse Trigonometric Functions**

1. Arcsine Function

$$\int \frac{1}{\sqrt{1-x^2}} dx = sin^{-1}(x) + c$$

2. Arccosine Function

$$\int \frac{-1}{\sqrt{1-x^2}} dx = \cos^{-1}(x) + c$$

3. Arctangent Function

$$\int \frac{1}{1+x^2} dx = \tan^{-1}(x) + c$$

4. Arccotangent Function

$$\int \frac{-1}{1+x^2} dx = \cot^{-1}(x) + c$$

5. Arcsecant Function

$$\int \frac{1}{x\sqrt{x^2-1}} dx = \sec^{-1}(x) + c$$

6. Arccosecant Function

$$\int \frac{-1}{x\sqrt{x^2-1}}dx = \operatorname{cosec}^{-1}(x) + c$$

## 4.3. Techniques for Integrating Trigonometric and Inverse Trigonometric Functions

Several techniques can be used to integrate trigonometric and inverse trigonometric functions, including substitution, integration by parts, and trigonometric identities.

### 4.3.1. Substitution

Substitution is a powerful technique that simplifies the integrand by changing variables. This method is particularly useful when integrating composite functions. We will be exploring this topic in the next subsection.

### 4.3.2. Integration by Parts

Integration by parts is useful when the integrand is a product of two functions. It is based on the product rule for differentiation. We will be exploring this topic in the next unit.

Integration of circular and inverse circular functions is an essential aspect of calculus, enabling the solution of various problems involving periodic phenomena and angles. By mastering the basic integrals and techniques such as substitution and integration by parts, one can effectively integrate these functions and apply them in different contexts. Understanding these concepts provides a strong foundation for further studies in mathematics and its applications in science and engineering.

# 5. INTEGRATION OF SIMPLE FUNCTIONS USING SUBSTITUTION

Integration by substitution is a powerful technique used to simplify the process of finding integrals, especially when dealing with composite functions. This method involves changing the variable of integration to make the integrand easier to handle. Below, we explore the concept of substitution, its theoretical foundation, and several examples to illustrate its application.

## 5.1. Understanding Substitution in Integration

Substitution in integration is based on the Chain Rule for differentiation. The idea is to transform a complicated integral into a simpler one by substituting a new variable.

If we have an integral of the form:

$$\int f\big(g(x)\big)g'(x)\,dx$$

We can make a substitution $u = g(x)$, which implies $du = g'(x)\,dx$. This transforms the integral into:

$$\int f(u)\,du$$

**Steps for Substitution**

1. Identify the substitution: Choose a substitution $u = g(x)$ that simplifies the integrand.
2. Differentiate the substitution: Find $du = g'(x)\,dx$
3. Transform the integral: Rewrite the integral in terms of $u$ and $du$.
4. Integrate: Perform the integration with respect to $u$.
5. Back-substitute: Replace $u$ with the original variable $x$.

**Examples**

Integrate $\int 2x\,cos(x^2)\,dx$

**Solution:**

1. Identify the substitution: Let

$$u = x^2$$

2. Differentiate the substitution:

$$du = 2x\,dx$$

3. Transform the integral:

$$\int (2x)\, cos(x^2)\, dx = \int cos(u)\, du$$

4. Integrate:

$$\int cos(u)\, du = sin(u) + c$$

5. Back-substitute:

$$sin(u) + c = sin(x^2) + c$$

So, $\int (2x)\, cos(x^2)\, dx = sin(x^2) + c$.

Integrate $\int x(x^2+1)^5\, dx$

**Solution:**

Let

$$u = x^2 + 1$$

Differentiate:

$$du = 2x\, dx$$

$$\frac{du}{2} = x\, dx$$

Transform the integral:

$$\int x(x^2+1)^5\, dx = \int (x\, dx)(u^5) = \int u^5 \frac{du}{2} = \frac{1}{2}\int u^5\, du$$

Integrate:

$$\frac{1}{2}\int u^5\, du = \frac{1}{2}\frac{u^6}{6} + c = \frac{u^6}{12} + c$$

Back Substitute:

$$\frac{u^6}{12} + c = \frac{(x^2+1)^6}{12} + c$$

Therefore,

$$\int x(x^2+1)^5\, dx = \frac{(x^2+1)^6}{12} + c$$

Integrate $\int sin^2(x) \cdot cos(x)$

**Solution:**

Let

$$u = sin\, x$$

Differentiate:

$$du = cos(x)\, dx$$

Transform the Integral:

$$\int sin^2(x)\, cos(x)\, dx = \int u^2\, du$$

Integrate:

$$\int u^2\, du = \frac{u^3}{3} + c$$

Back-substitute:

$$\frac{u^3}{3} + c = \frac{sin^3(x)}{3} + c$$

Therefore,

$$\int sin^2(x)\, cos(x)\, dx = \frac{sin^3(x)}{3} + c$$

Integrate $\int e^{2x}\, dx$

**Solution:**

Let,

$$u = 2x$$

Differentiate:

$$du = 2\, dx$$

$$dx = \frac{du}{2}$$

Transform the integral:

$$\int e^{2x}\, dx = \int e^u \frac{du}{2} = \frac{1}{2}\int e^u\, du$$

Integrate:

$$\frac{1}{2}\int e^u\, du = \frac{L}{2} e^u + c$$

Back-substitute:

$$\frac{1}{2} e^u + c = \frac{1}{2} e^{2x} + c$$

Therefore,

$$\int e^{2x}\, dx = \frac{1}{2} e^{2x} + c$$

Integration by substitution is a fundamental technique that simplifies the process of finding integrals, especially for composite functions. By changing variables, we can transform complicated integrals into simpler ones, making them easier to solve. Mastering this method is essential for tackling a wide range of problems in calculus and its applications in various fields.

INSPIRED BY LIFE

## 6. LIST OF INTEGRATION FORMULAS

1. Power Rule

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C \text{ for } n \neq -1$$

2. Exponential Functions

$$\int e^x dx = e^x + C$$

$$\int a^x dx = \frac{a^x}{\ln(a)} + C \text{ for } a > 0, a \neq 1$$

3. Logarithmic Functions

$$\int \frac{1}{x} dx = \ln|x| + C$$

4. Trigonometric Functions

$$\int \sin(x) dx = -\cos(x) + C$$

$$\int \cos(x) dx = \sin(x) + C$$

$$\int \sec^2(x) dx = \tan(x) + C$$

$$\int \csc^2(x) dx = -\cot(x) + C$$

$$\int \sec(x)\tan(x) dx = \sec(x) + C$$

$$\int \csc(x)\cot(x) dx = -\csc(x) + C$$

5. Inverse Trigonometric Functions

$$\int \frac{1}{\sqrt{1-x^2}} dx = \sin^{-1}(x) + C$$

$$\int \frac{-1}{\sqrt{1-x^2}} dx = \cos^{-1}(x) + C$$

$$\int \frac{1}{1+x^2} dx = \tan^{-1}(x) + C$$

$$\int \frac{-1}{1+x^2} dx = \cot^{-1}(x) + C$$

$$\int \frac{1}{|x|\sqrt{x^2-1}} dx = \sec^{-1}(x) + C$$

$$\int \frac{-1}{|x|\sqrt{x^2-1}} dx = \csc^{-1}(x) + C$$

6. Hyperbolic Functions

$$\int sinh(x)\,dx = cosh(x)\,dx + C$$

$$\int cosh(x)\,dx = sinh(x) + C$$

$$\int sech^2(x) = tanh(x) + C$$

$$\int cosech^2(x)\,dx = -\,coth(x) + C$$

$$\int sech(x)\,tanh(x)\;dx = -\,sech(x) + C$$

$$\int cosech(x)\,coth(x)\,dx = -\,cosech(x) + C$$

7. Inverse Hyperbolic Functions

$$\int \frac{1}{\sqrt{x^2+1}}dx = \sinh^{-1}(x) + C$$

$$\int \frac{1}{\sqrt{x^2-1}}dx = \cosh^{-1}(x) + C$$

$$\int \frac{1}{1-x^2}dx = \tanh^{-1}(x) + C$$

$$\int \frac{1}{1+x^2}dx = \coth^{-1}(x) + C$$

$$\int \frac{1}{x\sqrt{x^2-1}}dx = \mathrm{l}\,\mathrm{sech}^{-1}(\mathrm{x}) + C$$

$$\int \frac{1}{x\sqrt{x^2+1}}dx = \mathrm{cosech}^{-1}(\mathrm{x}) + C$$

8. Constant Rule

$$\int kdx = kx + C$$

## 7. SUMMARY

Alright, everyone, let's take a moment to reflect on what we've covered in this unit on Integration. We started by understanding the fundamental idea that integration is essentially the reverse of differentiation. It's not just about computing areas under curves; it's about accumulating quantities, which is crucial across various fields, from physics to economics.

We delved into the basic concepts of integration, starting with indefinite integrals, or antiderivatives, which are families of functions whose derivatives give back the original function. Remember, we add that constant $C$ because the derivative of a constant is zero, and we need to account for all possible antiderivatives.

Then, we moved on to definite integrals, which unlike indefinite integrals, give us a specific value representing the area under the curve between two points. This helps us quantify things like the total distance travelled by a car over a time interval given its velocity function, or the total quantity of a substance reacted or produced in chemical processes over time.

One of the highlights of this unit was tackling the Fundamental Theorem of Calculus. This theorem beautifully bridges the gap between differentiation and integration, showing that integrating a function over an interval and then differentiating the result brings us back to our original function. It's like taking a scenic route through the mathematical landscape and ending up right back where you started!

We also explored various techniques to tackle more complex integrals. We looked at substitution, which simplifies the integrand to make integration more manageable. Integration by parts, on the other hand, was our go-to method for functions that are products of two simpler functions—a direct counterpart to the product rule in differentiation.

Throughout this journey, we applied these concepts to solve practical problems, which not only grounded our understanding but also demonstrated the powerful applications of integration. Whether it was calculating the area of an irregular shape, determining the total change in a quantity, or even finding the work done in physics, integration stood out as a versatile and indispensable tool in our mathematical toolkit.

As we close this unit, remember that mastering integration is not just about solving equations—it's about understanding how things add up in the real world. So, take these

concepts, apply them, and always think about how you can use integration to make sense of the world around you. Let's gear up for the next unit, where we'll dive deeper into advanced techniques and applications of integration. Great job on getting this far, and I'm excited to see how you'll apply these concepts in even more complex scenarios!

INSPIRED BY LIFE

## 8. SELF-ASSESSMENT QUESTIONS

1. The process of finding the integral of a function is known as
2. What does the definite integral of a function $f(x)$ from $a$ to $b$ represent?
   A. The slope of the curve
   B. The derivative of the function
   C. The total accumulation of the quantity described by the function between $a$ and $b$
   D. The maximum value of the function between $a$ and $b$
3. The integral sign is denoted by the symbol
4. Which statement is true according to the Fundamental Theorem of Calculus?
   A. Integration and differentiation are unrelated.
   B. Differentiation is the inverse process of integration.
   C. Integration can be used to find the derivative of a function.
   D. The derivative of the integral of a function from $a$ to $x$ is the function itself.
5. If $F(x)$ is an antiderivative of $f(x)$, then $\int f(x)dx = F(x) +$ .
6. Which technique involves changing the variable of integration to simplify the integrand?
   A. Integration by parts
   B. Linear approximation
   C. Substitution
   D. Partial fractions
7. The definite integral $\int_a^b f(x)dx$ can be calculated using $F(b) - F(a)$, where $F$ is the of $f$.
8. Which of the following is an application of integration?
   A. Finding the area under a curve
   B. Determining the maximum point of a function
   C. Solving linear equations
   D. All of the above
9. The constant of integration is denoted by the letter _____.
10. Integration by parts is derived from which rule in differentiation?
    A. Chain rule

B. Product rule

C. Quotient rule

D. Power rule

INSPIRED BY LIFE

## 9. TERMINAL QUESTIONS

1. Calculate the indefinite integral of $f(x) = 3x^2 - 4x + 5$.
2. Evaluate the definite integral of $f(x) = x^3$ from $x = 0$to $x = 2$.
3. Using the substitution method, solve the integral $\int 2x\sin(x^2)dx$.
4. Find the area under the curve $f(x) = 4x$ from $x = 1$ to $x = 3$ using definite integration.
5. Apply integration by parts to solve $\int x\cos(x)dx$.
6. Calculate the integral of the exponential function $\int e^{3x}dx$..
7. Determine the integral of $\int \frac{dx}{x^2+4}$.
8. Using the power rule of integration, integrate $\int x^{-2}dx$ dx and discuss the condition of its application.
9. Evaluate the integral of the trigonometric function $\int \sec^2(x)dx$.
10. Compute the integral of the logarithmic function $\int \frac{1}{x}dx$ and discuss the implications for its domain.

## 10. ANSWERS

**Self-Assessment Questions**

1. Integration
2. C. The total accumulation of the quantity described by the function between $a$ and $b$
3. The integral sign
4. D. The derivative of the integral of a function from $a$ to $x$ is the function itself.
5. C
6. C. Substitution
7. Antiderivative
8. A. Finding the area under a curve
9. C
10. B. Product rule

**Terminal Questions**

1. $x^3 - 2x^2 + 5x + C$
2. $\left[\frac{x^4}{4}\right]_0^2 = \frac{2^4}{4} - \frac{0^4}{4} = 4$
3. $-\cos(x^2) + C$
4. $[2x^2]_1^3 = 2 \times 3^2 - 2 \times 1^2 = 18 - 2 = 16$
5. $x\sin(x) + \cos(x) + C$
6. $\frac{1}{3}e^{3x} + C$
7. $\frac{1}{2}\arctan\left(\frac{x}{2}\right) + C$
8. $-\frac{1}{x} + C$
9. $\tan(x) + C$
10. $\ln(|x|) + C$